

Data-Driven Analytics Framework for Demand Forecasting and Operational Optimization in Bike-Sharing Systems: Dublin Bikes Case Study

Shashank Ashok Gadavi (A00049544) and Prajwal Tejamurthy (A00050899)

MSc in Computing (Data Analytics)

Dublin City University, Ireland

Email: shashankashok.gadavi2@mail.dcu.ie and prajwal.tejamurthy2@mail.dcu.ie

Abstract—The availability of docks is crucial for a bike-sharing system, but most studies only look at it in parts. This paper focuses on dock availability at each station for Dublin Bikes, using thirteen months of detailed operational data (from August 2024 to August 2025, 7,416,360 timestamped observations across 115 stations). Instead of just focusing on one predictive task, we analyze eleven different aspects all in one process: short term demand forecasting, analyzing peak hours and seasonal trends, classifying stations that are almost empty or full, grouping stations based on behavior and time, detecting anomalies using statistics and models, recommending stations based on proximity, assessing geospatial coverage with Voronoi diagrams, analyzing operational KPIs (like utilisation, rebalancing, capacity planning, and uptime), breaking down time-series data, analyzing correlations between stations, and running Monte Carlo risk simulations. We use gradient-boosted tree models (XGBoost, LightGBM) that show excellent short-term forecasting accuracy ($R^2 = 0.967$ at the shortest horizon) and almost perfect risk discrimination for stations (AUC up to 0.998) without needing deep learning or GPU resources. The unsupervised methods reveal four different types of station behaviors and identify one malfunctioning station out of 115 from over seven million records, using just a simple physical-consistency check. We compare these results with a recent survey of bike-sharing research, which shows that infrastructure assessment, demand forecasting, and operations research are often done separately our pipeline helps bridge several gaps identified in the survey, using only open, simple, and reproducible methods, making it suitable for city wide deployment.

Index Terms—bike-sharing systems, demand forecasting, gradient boosting, station clustering, anomaly detection, Monte Carlo simulation, Voronoi analysis, smart-city analytics, urban mobility, rebalancing optimisation, cluster visualization, OpenStreetMap

I. INTRODUCTION

Bike-Sharing Systems (BSS) have become an integral part of multi-modal transport networks however, their daily operation ultimately depends on a simple but important issue keeping dock stations neither full nor empty. A rider typically abandons a system after two consecutive failed trips, and each empty or full station results in lost revenue (missed trips or stranded returns) or an unplanned truck delivery for rebalancing. Solving this well requires a holistic approach that combines short-term prediction with station behavior, network characteristics, and ongoing operational practice, rather than any single predictive model.

A current analysis of bike sharing and bike infrastructure research synthesized studies from a total of three categories an evaluation of open-data infrastructure quality, a forecast of future demand through a deep learning model, and the optimization of operational decisions and stated that there is little overlap between the three categories. Most of the infrastructure-quality evaluations do not use the forecasts developed in many of the demand-model papers for rebalancing and capacity decisions. In addition, according to the same study, almost all of the forecasting studies rely on large architectural models (e.g., hybrid LSTM (Long Short Term Memory) network, spatial-temporal graph convolutional network, Temporal Fusion Transformer), which require months of historical data and significant amounts of computing power. As such, this limits small systems with limited histories or recently established systems.

We have taken a completely different approach to this work. We created an integrated analytical pipeline for a Dublin Bike System using over 13 months of actual high frequency station status data. The pipeline includes (1) Forecasting (2) Classification (3) Clustering (4) Anomaly Detection (5) Geospatial Analysis and (6) Operational Decision Support on a Day-by-Day Basis. We are not developing or testing a new Deep Architecture. Our contributions in this study were Empirical and Integrative. Specifically, our empirical contribution was to provide evidence that established, well-known Gradient Boosting Methods as well as Classical Statistical Methods can be successfully applied across the entire life cycle of an operational bike sharing station with very little loss of accuracy. Further, we tied our results directly to the cross domain gaps identified by prior literature in terms of the need for integrating multiple domains. Additionally, where feasible, we cross validated each finding with an independent feature set developed from the same original dataset to determine if the same operational signals exist regardless of modeling paths.

Our primary contributions are as follows:

- **Integrated analytics pipeline:** A pipeline that was made up of eleven related tasks (forecasting, seasonal/peak trend analysis, risk classification, operational status classification, station and temporal clustering, anomaly / malfunction identification, a recommender, Voronoi coverage analysis, operational KPI analytics, decomposing a time

series for correlated stations by using cross-station correlations, and Monte Carlo simulation) evaluated against one dataset consisting of over 7 million actual records across 115 different stations.

- **Empirical evaluation of lightweight models:** Lightweight gradient-boosted tree models (XGBoost, LightGBM), using only lag/rolling/calendar features, reach strong short-horizon forecasting accuracy and near-perfect station-risk discrimination without deep learning or GPU infrastructure.
- **Geospatial-to-operational linkage:** A geospatial coverage-gap analysis (Voronoi partitioning), a behavioral station cluster, utilisation metrics and rebalancing recommendations are all directly linked by an integrated infrastructure operations gap identified in the earlier survey.
- **Interpretable, cross-validated malfunction detection:** The simple heuristic of bikes + free docks = capacity identifies one truly failing station among over seven million records without using manual labeling and that same station is also identified in a separate event based profile of trips as having the highest level of failure in that data set as well, giving additional confidence in the identification of that station's issue.

II. RELATED WORK

Open-data infrastructure assessment has moved from city-scale case studies toward national pipelines. Wysling and Purves fused OpenStreetMap (OSM) street networks with Paris planning-agency attributes to compute street-level bicycle suitability, validated against field audits of thirty streets [1]. Ferster et al. scaled a similar approach across fifteen Canadian cities using a standardised infrastructure classification scheme they report strong classification accuracy for protected infrastructure but also a persistent tagging ambiguity and an urban-rural completeness gap [2].

Demand forecasting has drifted from classical statistical and gradient-boosting models toward deep architectures that explicitly model spatial station interactions. Xu et al. proposed a multi-block hybrid deep-learning model that jointly leverages multiple data sources to predict bike-sharing supply and demand [3]. Chen et al. compared spatial-temporal graph convolutional variants for predicting availability in bike-sharing schemes and found that model performance is sensitive to how spatial dependencies between stations are represented [4]. Against this backdrop, the gradient-boosted tree ensembles we use XGBoost [14] and LightGBM [15], built on the classical random-forest idea of bagged decision trees [16] remain a strong, low-infrastructure baseline for tabular, feature-engineered time series, which is precisely the regime a single-city operator with a modest data science budget is in. This positioning is consistent with recent lightweight-ML forecasting work: Ramesh et al. demonstrated strong station-level demand prediction using classical ensemble methods rather than deep learning [11].

On the operations side, Cueva et al. reviewed ninety-six bicycle-sharing design studies and found that only a small fraction fold infrastructure-quality metrics into station-siting decisions [22]. Chang et al. combined demand forecasting with a data-driven rebalancing optimization framework for relocating operational and damaged bikes [10]. Talavera-Garcia et al. examined spatio-temporal mobility patterns in the BiciMAD (Madrid) bike-sharing system, illustrating how station-level usage patterns vary across different urban contexts [7]. Closer to the present study area, Yan et al. analyzed the parking behaviour of shared e-bike users in Dublin using a real-world operational dataset [6]. Across all studies the motivating survey reviewed, only two or three papers connect infrastructure quality to demand modelling, only two connect prediction to optimisation, and only one systematically addresses equity a persistent lack of integration across the three domains.

Finally, several of the unsupervised and statistical components used in Section IV draw on long-standing, well-validated methods rather than bespoke architectures: K-Means clustering [18], Ward-linkage agglomerative clustering, Isolation Forest anomaly detection [17], STL seasonal-trend decomposition [19], Voronoi spatial partitioning [20], and Monte Carlo stochastic simulation [21]. Using established, interpretable building blocks throughout rather than a single end-to-end deep model is a deliberate choice, consistent with the survey's observation that heavy architectures are often a poor match for the data volumes and technical capacity of small and mid-sized systems.

A. Research Objectives and Questions

Existing studies typically focus on only one component such as forecasting, optimization, or infrastructure assessment. Few studies integrate these analytical layers into a unified decision-support framework. The main goal of this project was to create an analytical data driven tool which can predict bike sharing usage and estimate bike share demand within Dublin using open source bike sharing data from stations and spatial data derived from Open Street Map (OSM).

This project will assess if there are any improvements to predictive outcomes when OSM data is incorporated as additional input layers for analysis. Furthermore, this project will provide insights for urban mobility planners to make informed decisions about the design and operation of bike-share systems.

The following questions guide the research:

1. What are the key factors (such as the day of the week, season, holidays and geospatial features extracted from OpenStreetMap) that influence bike demand and the number of trips made in Dublin?
2. Which machine learning models (such as, XGBoost, Random Forest, or Gradient Boosting) provide the most accurate and interpretable predictions for estimating bike trips from station-level data?
3. How does station-level geospatial context captured through Voronoi coverage analysis and behavioural clustering

relate to operational patterns such as utilisation, malfunction risk, and rebalancing need across the Dublin Bikes network?

These all questions attempt to resolve the issues of reliably predicting trip demand from aggregate station level data without knowing the origin and destination information. Additionally, identify how integrating spatial context may provide further insight into behavioral trends related to urban cycling in Dublin.

III. DATASET AND STUDY AREA

We used an extracted historical snapshot of data from all the available Dublin Bike stations that contain status information for each bike station regarding how many bikes were available to be rented, and how many empty slots existed for returning them, along with rental/return flag status, date/time when it was reported, capacity of station, and location of station (latitude and longitude). The raw snapshot contained approximately 7,416,360 rows that represent individual observations over time these observations were made at about a 10 minute interval. Each row represents an observation of one particular bike station. There are two derived measures of occupancy used throughout this analysis. utilisation is calculated as a percentage based on how much of a given station's total capacity is being utilized by available bikes.

Two derived occupancy metrics anchor the entire analysis. Utilisation is the fraction of a station's capacity currently occupied by bikes:

$$U = \frac{\text{bikes_available}}{\text{capacity}} \quad (1)$$

and its complement, is the fraction of free docks:

$$F = \frac{\text{docks_available}}{\text{capacity}} \quad (2)$$

with $U + F = 1$ by construction. A demand proxy, defined as $1 - U$, stands in wherever a single "how heavily is this station being used" signal is needed, on the reasoning that a drained station reflects recent pickup activity.

We also engineered calendar and temporal features from the report timestamp: hour, minute, day of week, weekend indicator, month, quarter, day/week of year, a five level times-of-day category (morning rush hours 7:00-10:00 AM, noon/midday, evening rush hours 4:00-8:00 PM, evening, nighttime), an Irish public-holiday flag, and a four season label. As a data quality warning we indicate that the holiday flag was generated using only the 2024 Irish public holiday calendar. Therefore, the holiday flags were created for the first eight months of the study window. The results should therefore reflect only part of the total effect of holidays on the classifications and forecasts.

A. Supplementary trip-event profiling

As another check to see if the utilisation based feature list was correct, we have also derived independently from the same data source, separate, trip event based on feature list. Dublin Bikes, represents the absolute number of bikes available at each station from one time period to the next thus, it is considered to be a single pick-up or drop-off

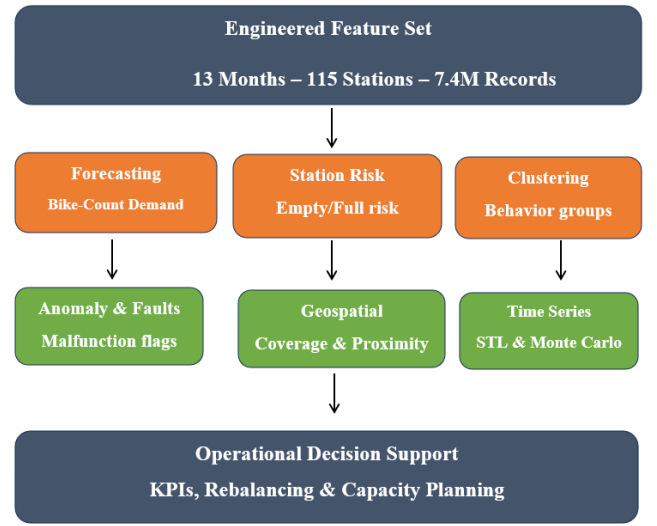


Fig. 1. End-To-End Pipeline Architecture of Dublin Bikes.

transaction, while daily trips aggregates all such transactions over 42,069 individual stations (and 41,264 after removing those with less than seven days of observations due to the lag required to create temporal features, including calendar and lag.) A per-station behaviour (mean daily trips, variation of trips to a station during a day, turnover-style Utilisation Rate = Daily_Trips/Capacity, percentage of days a station had zero or near-zero availability or near maximum capacity) is then used as an additional verification method of both clustering and malfunction detection in Section IV.

B. Analytical Pipeline Architecture

The data flow illustrated below is an end-to-end analytics process based on the engineered feature set (7.4 million data points from 115 stations over 13 months) and consists of three separate paths of modeling as follows:

Forecasting determines the number of bikes available at each station, providing input to the anomaly/fault detection module to identify malfunctioning docking stations or sensor units.

Risk assessment by station calculates the likelihood of an individual dock being full/empty, providing input to the geospatial analysis component to determine how far away stations are from other stations along with how well covered the entire network is.

Clustering categorizes stations based on their behavior patterns, providing input to time series analysis using STL decomposition and Monte Carlo simulations.

The results from these 6 modules provide output to the final decision support layer operational decision support where they are translated into actionable key performance indicators to guide rebalancing operations and capacity planning.

Therefore, while each major analytical path has its own value, it provides direct input to another analytical method down stream and all of them have a common purpose.

IV. METHODOLOGY

A. Short-Term Demand Forecasting

At Grand Canal Dock (Station 69), we have developed lag features for each of seven distances from the data point (1, 3, 6, 12, 24, 36, 48 rows in the table, approximately equivalent to 10 minutes through 8 hours of data), as well as three types of rolling averages a 1 hour moving average and its standard deviation, and a 4 hour moving average that are both calculated based upon lagged values so as to prevent leakage. Each type of feature was paired with time-based characteristics (hour, minute, weekday, weekend indicator, month and holiday indicator). The features were then used to fit three XGBoost regression models one for each forecasted time frame that utilized a chronological 80 / 20 partitioning scheme of the training set into test set with no shuffling to preserve the order of events. There were 200 decision trees within each model, the maximum tree depth was five levels and the learning rate was 0.05, random state 42 one per forecast horizon using a chronological 80/20 train-test split with no shuffling, to respect temporal ordering.

B. Peak-Hour and Long-Term Trend Analysis

We aggregated the average bike availability per station and hour-of-day was considered in order to determine the most active stations for the morning and evening peak periods (07:00-10:00 and 16:00-20:00, most active based on the lowest average bike availability at those times). Moving averages over 7 days and 30 days were used for the overall system daily average in order to distinguish between noise and seasonality effects.

C. Station Risk Classification

Two binary classification problems were formulated: whether or not the station would be almost empty (≤ 2 bikes) or almost full (≤ 2 available docks) about 30 minutes into the future (3 rows ahead). LightGBM models (with 200 estimators, learning rate 0.05, balanced classes, random state 42) were trained using nine features such as current number of bikes/docks, utilisation, hour, day of week, weekend indicator, month, holiday indicator and capacity intentionally omitting the lagged features, because the current state and the context seemed to contain enough information for the problem. The third model Random Forest which predicts whether the station stays in an active renting mode or not in the next snapshot.

D. Station and Temporal Clustering

All 115 stations were characterised by five measures: mean utilisation, variability of utilisation, a differential between mornings and evenings, weekends versus weekdays differential, and capacity. Standardization was followed by clustering based on K-Means the potential number of clusters varied in the range of $\{2, \dots, 8\}$ and was selected by means of inertia (or elbow criterion) and silhouette coefficient. While Silhouette coefficient achieved its maximum at $k = 2$ (≈ 0.328) and improved somewhat at $k = 3$ (≈ 0.300) and $k = 6$ (≈ 0.301), $k = 4$ (≈ 0.272) being a local minimum at approximately

0.272, we opted to choose $k = 4$, due to interpretability reasons, because not only did inertia curve bend in the neighborhood, $k = 4$ but those four clusters corresponded perfectly to the residential/office/tourist/low-utilisation classification, which can be used by the operational staff. Additionally, we considered mean 24-hour utilisation profile of each station to be a 24-dimensional vector and applied hierarchical Ward-linkage agglomerative clustering to group them in four temporal clusters.

E. Anomaly and Malfunction Detection

Anomalies of system-wide availability were detected with respect to system-wide hourly availability data by applying two separate approaches: one consisting of the rolling hourly (144 observations) z-score method with a threshold $|z| > 3$, and another approach utilizing the Isolation Forest algorithm (contamination ratio of 2%) applied to hourly series. On the station level, an anomaly can be indicated according to the physical consistency criteria in the case when the station has bikes and free docks approaching zero at the same time during a substantial portion of the observations ($\text{pct_zero_bikes} > 0.5$ or $\text{pct_full_docks} > 0.5$), or when a station is not in the active renting mode more than 30% of the time, or when a station has zero bike count variance ($\text{std} < 0.5$).

F. Recommender System

A proximity recommender that uses the haversine great-circle distance between the requesting station and all other stations was developed, where the stations were first filtered based on their capability to accommodate the request (≥ 2 bikes and actively renting, or ≥ 2 free docks and actively returning), returning the five nearest qualifying stations.

G. Geospatial Coverage Analysis

We built a Voronoi diagram over station coordinates using, which divides the entire city into regions of stations nearest to the service areas. The area of the cell is used as an indicator of the density of coverage of the area, while the cells that are very large or unbounded are marked as candidates for expansion.

H. Operational KPI Analytics

We conducted four types of operational studies: (1) The utilisation averages by station were ranked so we could identify overused or underused stations. (2) A rebalancing study was run where each station's excesses/shortages from a 50% utilisation target were identified. Stations with excess/shortage needs were identified in this manner. (3) An operational planning tool utilized a capacity planning rule where stations were flagged if they exceeded 80% usage greater than 40% of the time, and stations with less than 20% usage greater than 60% of the time would be considered potential candidates for expansion/reduction or relocation. (4) Uptime study analyzed average active percent for both renter and returner activity at each station utilizing 10 minute snapshots. As most users report data at similar intervals, it is estimated that a weighted-average utilisation measurement will approximate an availability measurement

TABLE I
XGBOOST SHORT-TERM FORECASTING PERFORMANCE (STATION 69
GRAND CANAL DOCK)

Horizon	MAE (bikes)	RMSE (bikes)	R ²
15 min	1.443	2.206	0.967
30 min	2.076	3.108	0.934
60 min	2.741	3.988	0.892

I. Advanced Time-Series Analysis

Using STL (Seasonal-Trend decomposition using Loess, robust mode), we used 24-hour periodicity on each of our 20 hourly time-series to separate each into its trend, daily seasonal components, and residual noise. In addition, we ran Pearson correlation on the hourly series from each station (there are 20 stations that have at least some data in every hour) as pairs of stations, to identify which of the 20 stations had similar movement in their occupancy.

J. Monte Carlo Risk Simulation

For Station 69, we modeled changes in the number of bikes available on an hour-by-hour basis using the samples from a normal distribution fit to the station's historic hourly differences (i.e mean μ and standard deviation σ). We generated 1,000 different 48-hour stochastic paths from the most recent measurement, clamped each path to be between [0, capacity=40] to obtain percentile bands (5th–95th) for the probability that the station will go below a nearly empty level or above a nearly full level at least once during the forecasted period.

V. RESULTS

A. Forecasting Accuracy

Forecast error rose and R², which is the expected behaviour for a lag-driven model (Table I). Fig. 2 compares actual and predicted bike counts over the final 200 test-set steps for all three horizons. This 15-minute-horizon accuracy (R²=0.967) is comparable to or exceeds the short-term availability accuracy reported for deep Bi-LSTM approaches on similar bike-sharing station data [9], and is consistent with other work showing that lightweight ensemble models can match deep-learning-level accuracy for station-level demand prediction without GPU infrastructure [11].

Grand Canal Dock (Station 69) later determined to be the sole malfunctioning station identified in Section V, also represented the single busiest station at peak rush hour measured by this metric, on average having only 1.12 bikes available for the 7:00-9:00 am / 4 pm-7 pm time frame. In retrospect, it's reasonable to conclude that this represents a station that had rarely reported an accurate bike count due to a malfunctioning sensor versus a station under unusually high legitimate usage.

System-wide daily average availability, smoothed with 7- and 30-day moving averages, together with the month-by-month distribution of station-level averages, are shown in Fig. 3 the weekday-versus-weekend hourly comparison is

TABLE II
CLASSIFICATION PERFORMANCE (*SEVERE CLASS
IMBALANCE—SUPPORT 1,561 VS. 1,481,688)

Task	AUC	Prec. (At-Risk)	Recall (At-Risk)
Near-empty risk (30 min)	0.990	0.84	0.97
Near-full risk (30 min)	0.998	0.77	0.98
Operational status (t+1)*	0.785	0.17	0.38

shown in Fig. 4. Weekday profiles show a pronounced double dip around the morning and evening commute, while weekend profiles are flatter and shifted later in the day, consistent with leisure rather than commute-driven use.

B. Risk Classification

Both the full-saturation-risk classifier and the empty-saturation-risk classifier both had very good discrimination (Table II). The near-full model did slightly better than the near-empty model. This makes sense because near-full events are likely to be influenced by structural factors, i.e., the fact that most people travel to a specific area (such as an office or public transportation), rather than the fact that riders are making individual choices for near-empty events. This use of current-state and contextual features rather than lagged time-series features for station-level risk prediction is consistent with prior work using Random Forest models driven by station activity [12], and the strong discrimination achieved here (AUC up to 0.998) supports the broader case for interpretable tree-based classifiers over black-box alternatives, echoing recent explainable-AI approaches to bike-sharing station demand [8]. The operational-status classifier had a respectable AUC (0.785), however it has little real-world utility at this time. "Not Renting" images made up only 1,561 of 1,483,249 test set records therefore the precision for that class (0.17), although higher than the recall, was still relatively low.

C. Station Clustering

Fig. 5 represents the inertia and silhouette that were used to determine k. The four clusters identified in Table III demonstrate clean separation between each of the clusters based on the differences between rush hour and weekday/weekend usage. For example, Cluster 3 has a large positive rush differential (9.878) showing it is morning heavy. This means that commuters originate from this station during peak hours, and it drains ridership in the mornings and then fills up at night. On the other hand, cluster one shows both the highest average utilisation (0.542) along with the lowest average capacity (27.97) and the largest positive difference in weekend minus weekday (4.35). These results indicate small, highly utilized, and also predominantly leisure-leaning stations. Finally, the two remaining clusters have zero or negative rush differentials while displaying low levels of utilisation (0.188 and 0.377). Therefore these results illustrate underutilized or primarily destination/office-type behavior.

Ward-linkage agglomerative clustering of each station's 24-hour utilisation profile dendrogram, mean cluster profiles)

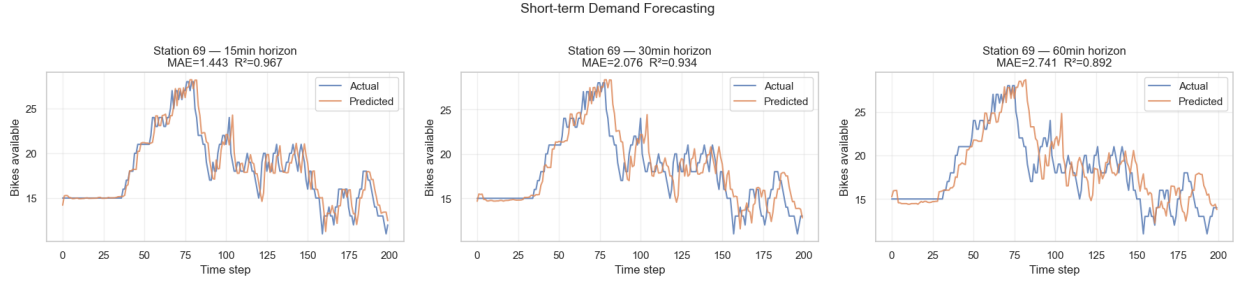


Fig. 2. Actual vs. predicted bikes available, station 69, for the three forecast horizons.

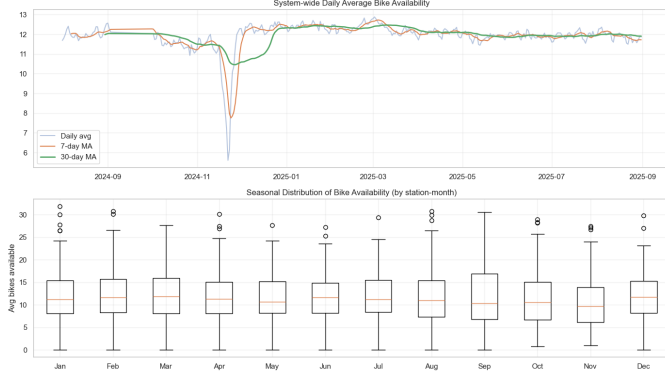


Fig. 3. System-wide daily average bike availability with 7-day and 30-day moving averages (top), and seasonal distribution of station-month average availability (bottom).

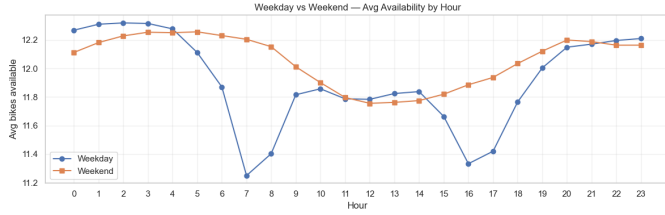


Fig. 4. Weekday vs. weekend average bike availability by hour of day, system-wide.

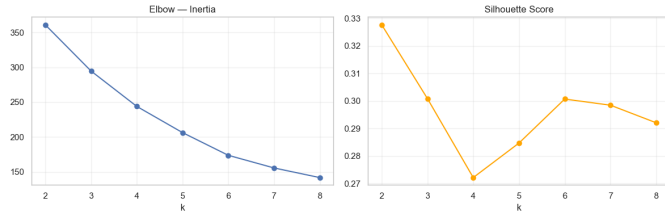


Fig. 5. K-Means elbow (inertia, left) and silhouette score (right) across candidate cluster counts $k = 2 \dots 8$.

largely echoes this structure from a purely temporal-shape perspective: the tallest merge in the dendrogram (height ≈ 10.5) splits the network into two broad daily-rhythm families, which the four-cluster cut further divides. One temporal cluster is markedly more distinctive than the others an inverted daily curve, high overnight and drained by early morning a strong

TABLE III
K-MEANS CLUSTER PROFILES ($k = 4$)

Cl.	Avg. Util.	Std. Util.	Rush D.	Wknd–Wkday	Cap.
0	0.188	0.180	0.58	−1.27	33.83
1	0.542	0.300	−2.73	+4.35	27.97
2	0.377	0.265	−2.66	−1.48	32.31
3	0.305	0.270	9.88	−4.19	37.25

signature of a residential, commuter-origin station consistent with Cluster 3 above.

D. Anomaly and Malfunction Detection

The rolling z-score method flagged 1,270 anomalous hours system-wide, versus 2,085 for Isolation Forest (Fig. 6, Table VI); the gap reflects the z-score method’s lack of an upper bound on deviation magnitude, unlike Isolation Forest’s fixed contamination cutoff. Both methods show similar patterns overall. At the station level, exactly one of 115 stations York Street West, Station 51 was flagged as anomalous: 90.2% of its records showed zero bikes available and 89.4% showed zero free docks available simultaneously (Fig. 7), a combination that is physically impossible under normal operation. This is independently corroborated by the trip-event profiling in Section III: using a $>25\%$ nearly-empty/nearly-full day threshold, Station 51 again ranks highest network-wide, with 92.9% of its days classified as nearly empty consistent with the utilisation-based 90.2% figure. Nineteen other stations passed this looser trip-based criterion (Table IV) but not the stricter utilisation-based one, since each shows normal, non-zero variation in daily trip counts unlike Station 51, which reports essentially no usable data on almost all days.

This diagnosis is supported in addition by the trip-event profiling finding (Section III). When a station is defined as either nearly empty or nearly full on the basis of trip count (i.e., more than 25% of its days are classified as such), then Station 51 once again ranks as the highest-ranking station in terms of extremes within the network. In particular, based on the trip-count classification method, Station 51 was found to have 92.9% of all days at Station 51 identified as nearly empty, which corresponds to and exceeds the 90.2% nearly-empty classification rate obtained using the utilisation-based approach. A total of nineteen other stations were identified



Fig. 6. System-wide hourly anomalies: rolling 24-hour z-score, $|z| > 3$ (top) Isolation Forest, 2% contamination (bottom).

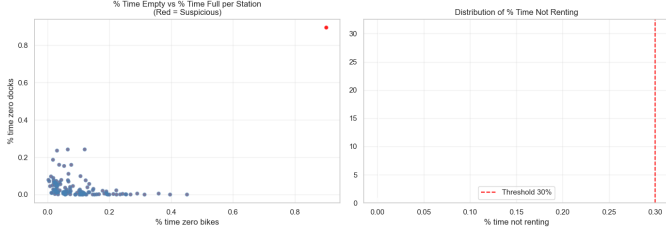


Fig. 7. % time with zero bikes vs. % time with zero free docks per station, coloured by suspicious flag (left) distribution of % time not actively renting, with the 30% decision threshold (right).

TABLE IV
TOP 10 OF 20 STATIONS FLAGGED BY THE INDEPENDENT TRIP-BASED CRITERION ($>25\%$ OF DAYS NEARLY EMPTY OR NEARLY FULL)

Station ID	Name	Empty rate
51	York Street West	0.929
59	Denmark Street Great	0.742
61	Hardwicke Place	0.645
89	Fitzwilliam Sq. East	0.647
30	Parnell Square North	0.592
79	Eccles Street East	0.478
111	Mountjoy Square East	0.505

as having passed the lower, day-level trip-criteria used here however, none of those nineteen stations would have been identified via use of the more restrictive utilisation-based approach. Those nineteen additional stations (Table IV), could be expected to be identified since they do not exhibit the chronic imbalance characteristics associated with Station 51, instead each of the nineteen stations exhibits a reasonable non-zero standard deviation in daily trip counts thus indicating there is activity at each of these stations that may be considered normal/acceptable whereas, Station 51 has essentially no data reported during almost all days.

E. Geospatial Coverage and Recommender

The Voronoi partition (Fig 8) clearly illustrates the spatial pattern of cell size in Dublin. At the center of the city (i.e. at the core) there are many very small, closely-packed cells. Moving away from this center toward the north or west, the cells increase in size until they reach their maximum size at

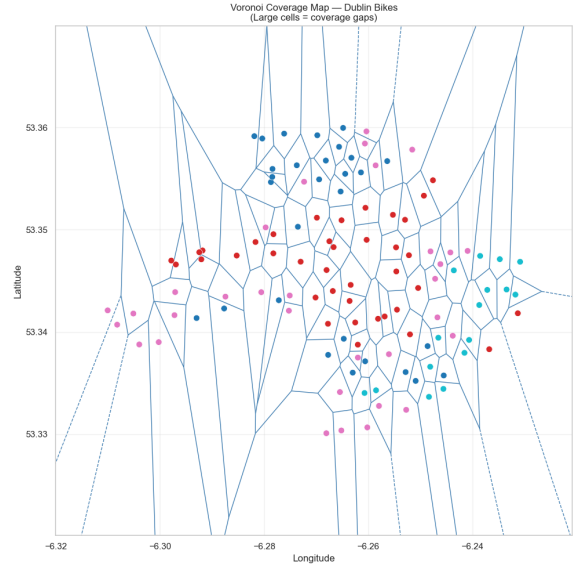


Fig. 8. Voronoi coverage diagram over the 115 Dublin Bikes stations, points coloured by K-Means behavioural cluster.

the edges of the network (e.g., the Docklands). This suggests that these periphery areas may be appropriate candidates for future development, consistent with GIS-based approaches that use spatial-integration metrics to flag infrastructure gaps around bike-sharing stations [13]. Visually, the largest Voronoi cells correspond to Station Clusters 0/2 (low utilisation) as identified in Section V, thus providing some validation to our findings regarding both the geographic characteristics of the network and how people use it. For example, when we perform a representative search for nearby stations from station 1 (Clarendon Row), we find York Street East (243.5 m, 18 bikes), Golden Lane (347.6 m, 9 bikes), Frederick Street South (380.6 m, 22 bikes), St. Stephen's Green South (383.2 m, 7 bikes) and Fownes Street Upper (412.9 m, 16 bikes) as the top five potential locations to pick up a bike.

F. Operational Analytics

At some locations, utilisation rates are far higher (over 80%) and at other locations very low (less than 20%), with many locations being somewhere in between. The rebalancing model showed there were 261 excess bicycles on 35 bicycle parking stations when compared to a shortage of 732 bicycles on 75 bicycle parking stations, which indicates that approximately half of all bicycle parking stations have bicycles available for rent. Based upon capacity-planning rules, it was determined that 81 percent (93/115) of all bicycle parking stations were appropriately sized (Table V), which has been utilized to estimate the minimum amount of space required to meet future demand. Uptime analysis indicated that none of the stations had an uptime of less than 80%, therefore any operational issues within this network appear to be long term imbalances and not the lack of service due to unavailability. This surplus/deficit framing is conceptually aligned with optimization-based rebalancing frameworks in the literature

TABLE V
CAPACITY-PLANNING CLASSIFICATION (115 STATIONS TOTAL)

Recommendation	Number of Stations
OK (no action)	93
Reduce / relocate	17
Expand capacity	5

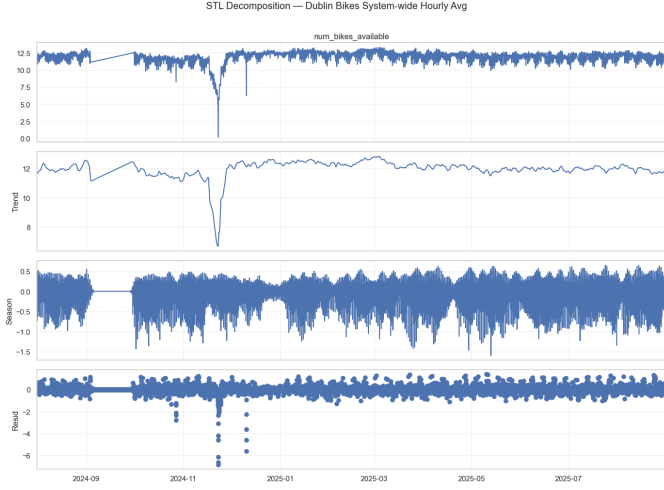


Fig. 9. STL decomposition of the system-wide hourly average bike availability series (observed, trend, seasonal, residual).

[5], [10], although unlike those approaches ours is a descriptive capacity-planning classification rather than a formal routing or scheduling optimization.

G. Time-Series Decomposition and Cross-Correlation

STL decomposition of the system-wide hourly series (24-hour period, Fig. 9) produced a trend component ranging from 6.67 to 12.84 bikes, a seasonal (daily) component ranging from -1.59 to 0.65 bikes, and a residual standard deviation of 0.282 a small residual relative to the trend range, indicating the 24-hour periodicity captures most of the structured variation in system-wide availability. Among the twenty stations with the most complete records, the strongest pairwise correlation came in at 0.832 (stations 20 and 19), which suggests shared catchment areas or correlated commuter flows, a pattern also reported in spatio-temporal analyses of other bike-sharing networks such as BiciMAD in Madrid [7].

H. Monte Carlo Simulation

Over a 48 hour time frame (See Fig. 10), the Monte Carlo Simulation estimated that there was a 62.4% chance that Station 69 will fall below the near empty level at some point during this period versus a 35.3% chance that it will be near or above the near full level. This asymmetry indicates the system is approximately 2 times as likely to "run out" as to "overflow". This type of risk exposure cannot be captured with a single point estimate and can only be determined by simulating the entire distribution of possible future states of trajectories instead of just an individual "expected" state. This stochastic

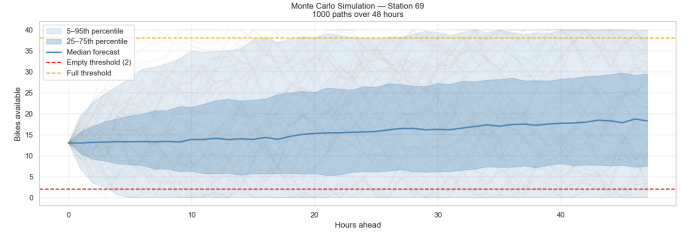


Fig. 10. Monte Carlo simulation of 48-hour bike availability, station 69 (1,000 simulated paths 5th95th and 25th75th percentile bands shown).

TABLE VI
SUMMARY OF ANOMALY AND MALFUNCTION DETECTION ACROSS METHODS

Method	Anomalies / Flags
Rolling 24h z-score ($ z > 3$)	1,270 hourly points
Isolation Forest (2% contam.)	2,085 hourly points
Station-level physical-consistency	1 station (of 115)
Trip-based day-level criterion	20 stations (of 115)

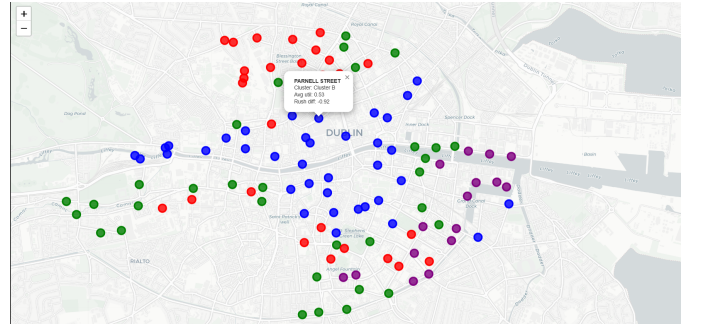


Fig. 11. Interactive OpenStreetMap visualization of K-Means clustering for the Dublin Bikes network.

framing complements deterministic rebalancing frameworks in the literature [10], which typically forecast a single expected demand trajectory rather than a full risk distribution.

I. OpenStreetMap-Based Cluster Visualization

An interactive map of the Dublin Bikes stations and their K-Means clusters was built using the Folium Python library as an interface to OpenStreetMap. Each station is plotted at its geographic coordinates (latitude/longitude) as a colour-coded marker corresponding to its K-Means cluster (Clusters AD), with interactive pop-ups showing the station name, cluster assignment, average daily utilisation, and rush-hour usage differential. Combining the clustering results with this geospatial visualisation provides an accessible tool for investigating the geography of station behaviour and supporting operational decisions such as rebalancing and infrastructure planning.

J. OpenStreetMap-Based Proximity Recommendation

To enhance the usability of the Dublin Bikes system, a proximity-based recommendation module utilizing OpenStreetMap (OSM) data was created. This module helps users

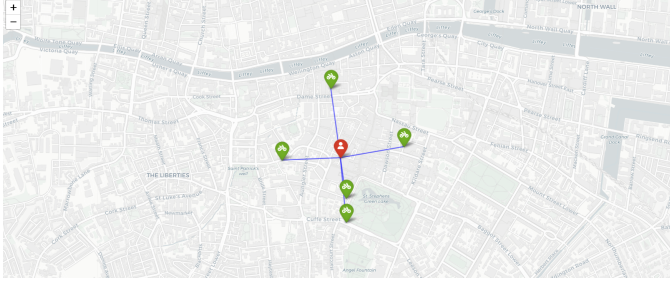


Fig. 12. OpenStreetMap visualization of the proposed proximity recommendation system.

determine which alternative bike stations have sufficient bicycles available for rent/pickup or provide adequate space for return/docking by determining which of the nearest stations meet these criteria.

Firstly, all bike stations are filtered to identify those currently operating and having the required number of bicycles for rental/pickup, or docks for return. Those meeting the requested criteria (for instance having at least 2 bicycles available for pick-up or at least 2 free docking spaces for return), become eligible candidate stations for the recommendation. Next, using the Haversine Distance Formula, the distances from each candidate station to the original selected station are calculated. Lastly, the five nearest qualifying candidates are presented as recommendations.

A graphic illustration (OpenStreetMap) is shown below (Fig. 12). The Red Marker represents the user-selected bike station. The Green Markers are the candidate bike stations which met the requirements of the user and were identified as being close enough to walk to. Blue Lines connect the selected station to each of the suggested candidate stations illustrating the most direct paths geographically between each selected station and its alternatives.

The visual illustration clearly presents how the location of bike stations can be used to assist users make informed decisions based upon proximity to other bike stations along with up-to-date information regarding the number of available bikes or empty docking stations. The new method utilizes both geographic proximity and station availability, thus enabling a reduction in travel distance and/or an increased probability of acquiring/returning a rented bicycle.

VI. DISCUSSION: POSITIONING AGAINST THE LITERATURE

Table VII maps each module of the pipeline described in Section IV against the cross-domain gaps identified by the motivating survey in Section II. In contrast to much of the surveyed literature that treats infrastructure assessment, demand forecasting, and operations as separate areas of study, the workflow here ties all three together in a continuous process at the expense of some possible peak performance using an inexpensive, well-understood pipeline based on classical geometric/statistical methods and gradient-boosted trees. The Station 51 malfunction diagnosis (Section V), independently

TABLE VII
POSITIONING AGAINST CROSS-DOMAIN GAPS IDENTIFIED IN THE SURVEYED LITERATURE

Gap (per survey)	Status in Prior Work	This Pipeline
Infrastructure–demand linkage	Addressed in ≤ 3 of 12 papers	Correlates Voronoi density with cluster/utilisation KPIs
High data/compute needs	Deep models, ≥ 6 mo. data, GPU	Gradient boosting, standard CPU
Prediction–optimisation link	Addressed in 2 of 12 papers	Monte Carlo bands feed rebalancing decisions
Single-metric validation	Mostly single-metric	Eleven analytical layers, 2 independent feature sets

confirmed by two separately engineered feature sets, illustrates the value of this triangulation: when multiple analyses performed separately on the same data reach the same conclusion, the resulting operational signal is more reliable than the output of any single model and such triangulation is achievable within a lightweight pipeline in a way that is harder to replicate with a single large-scale, end-to-end model.

VII. LIMITATIONS

Several limitations qualify our assessment:

- **Single-station demonstration:** The forecasting and Monte Carlo simulation models were demonstrated on only one high-traffic station out of 115.
- **Class imbalance:** The operational-status classifier has severe class imbalance (support 1,561 vs. 1,481,688), limiting its current operational reliability.
- **Missing external covariates:** Weather, calendar events, and public-transport disruptions were not included, though the literature identifies these as differentiating factors.
- **Incomplete holiday calendar:** The holiday flag covers only 2024, so any holiday effect in the models reflects part of the study window only.
- **Non-optimized hyperparameters:** Cluster count ($k = 4$) and risk thresholds were chosen for interpretability rather than formal optimization $k = 4$ trades a lower silhouette score for operational clarity versus the statistically optimal $k = 2$ or $k = 6$.
- **Scope:** The study covers a single city over a thirteen-month period.

VIII. FUTURE WORK

A few natural extensions stand out joint, multi-station forecasting (a shared gradient-boosted model with station embeddings, or a lightweight graph-based model) to test whether the cross-station correlations identified in Section V improve on the single-station baseline folding in weather and public-holiday-adjacent event data, including completing the holiday

calendar through 2025 resampling, or cost-sensitive learning to address the operational-status class imbalance and a formal, uncertainty-aware rebalancing optimization that consumes the Monte Carlo risk bands directly rather than the point-estimate surplus/deficit model used here.

IX. CONCLUSION

The findings of this study indicate that an entire analytical pipeline consisting of lightweight gradient-boosted tree-based algorithms, classical clustering techniques, basic statistical anomaly detection methods and elementary geospatial and stochastic simulation methods may be used to cover all stages of the operational life cycle of a bike-share station. This was demonstrated using thirteen months of actual usage data for Dublin Bikes, with the pipeline yielding high levels of accuracy (R^2 up to 0.967 and AUC up to 0.998) as well as directly addressing identified gaps in the integration of physical infrastructure, demand and operational needs by a recent survey found were unresolved across much of the larger body of bike-sharing research. An independent second feature set, designed and implemented separately from the first, further validated the primary diagnostic result of the pipeline (i.e., identifying one malfunctioning station), providing additional support for the results presented in the absence of any manual labeling. Lastly, both the process and computational requirements have been intentionally made as simple and reproducible as possible so that they could serve as a useful example/prototype for other small cities or newly deployed stations.

X. AUTHOR DISCLAIMER

We acknowledge that limited assistance from generative AI tools was used during the writing process specifically to improve phrasing and fluency in a few sections, such as the abstract, future work and conclusion, where expressing ideas concisely was particularly important. This support was applied sparingly and only for language refinement. All research content, technical methods, analysis, and interpretations are entirely our own.

REFERENCES

- [1] L. Wysling and R. S. Purves, "Where to improve cycling infrastructure? Assessing bicycle suitability and bikeability with open data in the city of Paris," *Transportation Research Interdisciplinary Perspectives*, vol. 15, p. 100648, 2022. Available: <https://doi.org/10.1016/j.trip.2022.100648>
- [2] C. Ferster, T. Nelson, K. Manaugh, J. Beairisto, K. Laberee, and M. Winters, "Developing a national dataset of bicycle infrastructure for Canada using open data sources," *Environment and Planning B: Urban Analytics and City Science*, vol. 50, no. 9, pp. 25432559, 2023. Available: <https://doi.org/10.1177/23998083231159905>
- [3] M. Xu, H. Liu, and H. Yang, "A deep learning based multi-block hybrid model for bike-sharing supply-demand prediction," *IEEE Access*, vol. 8, pp. 8582685838, 2020. Available: <https://doi.org/10.1109/ACCESS.2020.2987934>
- [4] Z. Chen, H. Wu, N. E. O'Connor, and M. Liu, "A comparative study of using spatial-temporal graph convolutional networks for predicting availability in bike sharing schemes," in *Proc. IEEE Int. Intelligent Transportation Systems Conf. (ITSC)*, Indianapolis, USA, Sep. 2021, pp. 12991305. Available: <https://doi.org/10.1109/ITSC48978.2021.9564831>
- [5] R. Hu, Z. Zhang, X. Ma, and Y. Jin, "Dynamic Rebalancing Optimization for Bike-Sharing System Using Priority-Based MOEA/D Algorithm," *IEEE Access*, vol. 9, pp. 2706727084, 2021. Available: <https://doi.org/10.1109/ACCESS.2021.3058013>
- [6] S. Yan, M. Liu, and N. E. O'Connor, "Parking Behaviour Analysis of Shared E-Bike Users Based on a Real-World Dataset - A Case Study in Dublin, Ireland," in *Proc. IEEE 95th Vehicular Technology Conference (VTC2022-Spring)*, Helsinki, Finland, 2022. Available: <https://doi.org/10.1109/VTC2022-Spring54318.2022.9860871>
- [7] R. Talavera-Garcia, G. Romanillos, and D. Arias-Molinares, "Examining Spatio-Temporal Mobility Patterns of Bike-Sharing Systems: The Case of BiciMAD (Madrid)," *Journal of Maps*, vol. 17, no. 1, pp. 713, 2021. Available: <https://doi.org/10.1080/17445647.2020.1866697>
- [8] F. Ngeni, B. Kutela, T. J. Chengula, C. Ruseruka, H. Musau, N. Novat, D. A. Indah, and S. Kasomi, "Prediction of bike-sharing station demand using explainable artificial intelligence," *Machine Learning with Applications*, vol. 17, p. 100582, 2024. Available: <https://doi.org/10.1016/j.mlwa.2024.100582>
- [9] E. Collini, P. Nesi, and G. Pantaleo, "Deep learning for short-term prediction of available bikes on bike-sharing stations," *IEEE Access*, vol. 9, pp. 124337124347, 2021. Available: <https://doi.org/10.1109/ACCESS.2021.3110794>
- [10] X. Chang, J. Wu, H. Sun, G. H. de Almeida Correia, and J. Chen, "Relocating operational and damaged bikes in free-floating systems: A data-driven modeling framework for level of service enhancement," *Transportation Research Part A: Policy and Practice*, vol. 153, pp. 235260, 2021. Available: <https://doi.org/10.1016/j.tra.2021.09.010>
- [11] A. A. Ramesh, S. P. Nagiseti, N. Sridhar, K. Avery, and D. Bein, "Station-level demand prediction for bike-sharing system," in *Proc. 2021 IEEE 11th Annual Computing and Communication Workshop and Conference (CCWC)*, 2021. Available: <https://doi.org/10.1109/CCWC51732.2021.9375958>
- [12] Y.-H. Seo, S. Yoon, D.-K. Kim, S.-Y. Kho, and J. Hwang, "Predicting demand for a bike-sharing system with station activity based on random forest," *Proceedings of the Institution of Civil Engineers Municipal Engineer*, vol. 174, pp. 97107, 2021. Available: <https://doi.org/10.1680/jmuen.20.00001>
- [13] M. J. Kłos and P. Soczówka, "A GIS-based method of the assessment of spatial integration of bike-sharing stations," *Sustainability*, vol. 13, p. 3894, 2021. Available: <https://doi.org/10.3390/su13073894>
- [14] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, San Francisco, CA, USA, Aug. 2016, pp. 785794.
- [15] G. Ke et al., "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems 30 (NIPS 2017)*, Long Beach, CA, USA, Dec. 2017, pp. 31463154.
- [16] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 532, 2001.
- [17] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *Proc. 8th IEEE Int. Conf. Data Mining (ICDM)*, Pisa, Italy, 2008, pp. 413422.
- [18] J. MacQueen, "Some methods for classification and analysis of multivariate observations," in *Proc. 5th Berkeley Symp. Mathematical Statistics and Probability*, vol. 1, 1967, pp. 281297.
- [19] R. B. Cleveland, W. S. Cleveland, J. E. McRae, and I. Terpenning, "STL: A seasonal-trend decomposition procedure based on loess," *Journal of Official Statistics*, vol. 6, no. 1, pp. 373, 1990.
- [20] F. Aurenhammer, "Voronoi diagrams a survey of a fundamental geometric data structure," *ACM Computing Surveys*, vol. 23, no. 3, pp. 345405, 1991.
- [21] N. Metropolis and S. Ulam, "The Monte Carlo method," *Journal of the American Statistical Association*, vol. 44, no. 247, pp. 335341, 1949.
- [22] F. Cueva, P. Shi, and P. Cedillo, "Designing Bike-Sharing Systems Supported by Data: A Systematic Literature Review," *IEEE Access*, vol. 12, pp. 162731162754, 2024. Available: <https://doi.org/10.1109/ACCESS.2023.3337235>